



SELECTION SORT

DAA

DESIGN AND
ANALYSIS OF
ALGORITHMS

★★★

A SIMPLE AND EFFICIENT SORTING ALGORITHM



Selection Sort is a comparison-based sorting algorithm. It repeatedly selects the minimum element from the unsorted part and places it at the beginning of the unsorted part.

HOW IT WORKS



1. Find the minimum element in the unsorted part of the array.



2. Swap it with the first element of the unsorted part.



3. Move the boundary of unsorted part one position forward.



4. Repeat the above steps until the entire array is sorted.

ALGORITHM (PSEUDOCODE)

```
SelectionSort(A[0..n-1])
  for i ← 0 to n-2
    min ← i
    for j ← i+1 to n-1
      if A[j] < A[min]
        min ← j
    if min ≠ i
      swap A[i] and A[min]
```

EXAMPLE

Consider the array:

64	25	12	22	11
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Pass 1: Find minimum (11) and swap with first element.

11	25	12	22	64
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Pass 2: Find minimum (12) in remaining and swap.

11	12	25	22	64
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Pass 3: Find minimum (22) in remaining and swap.

11	12	22	25	64
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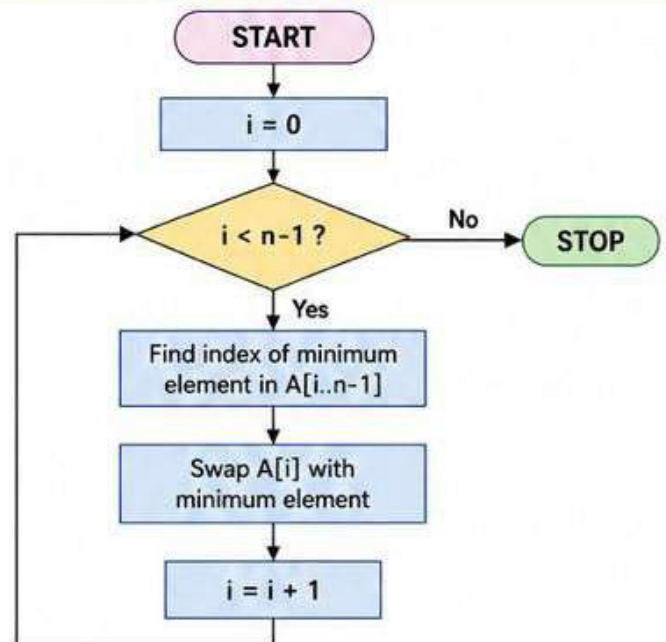
Pass 4: Find minimum (25) in remaining and swap.

11	12	22	25	64
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Sorted Array:

11	12	22	25	64
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FLOWCHART



KEY POINTS

- ★ In-place sorting algorithm (uses constant extra space).
- ★ Not a stable sorting algorithm.
- ★ Performs better on small datasets.
- ★ Simple to understand and implement.



COMPLEXITY ANALYSIS

Best Case	$O(n^2)$
Average Case	$O(n^2)$
Worst Case	$O(n^2)$
Space Complexity	$O(1)$



PROS & CONS

✓ PROS

- Simple and easy to implement
- Requires only constant extra space

✗ CONS

- Not efficient for large datasets
- Time complexity is $O(n^2)$ in all cases



“ The simplest algorithms are often the building blocks of the most efficient solutions. ”

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